

Fog Lamps

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST3093-A	Fluke 77-1V Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

The Steering Column Control Module (SCCM) monitors the fog lamp switch (integrated into the headlamp switch) and sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) , indicating the fog lamp switch position.

The fog lamps can be turned on when the following conditions are met:

- The ignition is in RUN or START
- The low beam headlamps or the parking lamps are on
- The high beams are off

When the BCM receives an input from the headlamp switch indicating a request for the fog lamps, the BCM provides ground to the fog lamp relay (integrated into the BCM). When the fog lamp relay is energized, voltage is routed to the fog lamps.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of electrical damage.

Visual Inspection Chart

Electrical

- Body Control Module (BCM) fuse 47 (15A) (fog lamps)
- Bulb
- Wiring, terminals or connectors
- Headlamp switch

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:

- verify the ignition is on.
 - The air bag warning indicator prove-out confirms ignition on (other indicators may not prove ignition on).
If the ignition does not turn on, refer to Section 211-05 to diagnose no power in run.
- verify the scan tool operation with a known good vehicle.
- refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

7. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
- If the network test passes, retrieve and record the continuous memory DTCs.

8. Clear the continuous DTCs and carry out the self-test diagnostics for the BCM and the SCCM .

9. If the DTCs retrieved are related to the concern, refer to Diagnostic Trouble Code (DTC) Chart in this section. For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

Symptom Chart

Symptom Chart

Condition	Possible Causes	Action
• Both fog lamps are inoperative	• Fuse • Wiring, terminals or connectors • Headlamp switch • Steering Column Control Module (SCCM)	• VERIFY the bulbs are OK. If OK, GO to Pinpoint Test Q.

	Body Control Module (BCM)	
An individual fog lamp is inoperative	Wiring, terminals or connectors	VERIFY the bulb is OK. If OK, GO to Pinpoint Test R.
The fog lamps are on continuously	- Wiring, terminals or connectors - Headlamp switch - SCCM - BCM	GO to Pinpoint Test S.
The fog lamps on indicator is inoperative	- Wiring, terminals or connectors - Headlamp switch - BCM	GO to Pinpoint Test T.

Pinpoint Tests

Pinpoint Test Q: Both Fog Lamps Are Inoperative

Refer to Wiring Diagrams Cell [86](#) , Fog Lamps for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) monitors the fog lamp switch position by sending a voltage signal to the headlamp switch. When the fog lamp switch is engaged, the voltage signal is routed to ground indicating a request for the fog lamps.

The SCCM sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) , indicating the fog lamp switch position. When the BCM receives input from the SCCM indicating a request for the fog lamps, the BCM provides ground to the fog lamp relay (integrated into the BCM). When the fog lamp relay is energized, voltage is routed to the fog lamps.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Headlamp switch
- SCCM
- BCM

PINPOINT TEST Q : BOTH FOG LAMPS ARE INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
Q1 CHECK THE FOG LAMPS PID
- Ignition ON. - Enter the following diagnostic mode on the scan tool: BCM DataLogger . - Monitor the BCM fog lamps PID (FOG_LAMPS) while engaging the fog lamp switch.

Does the PID indicate the fog lamps are on?

Yes	VERIFY the BCM fuse 47 (15A) is OK. If OK, GO to Q5 . If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short.
No	GO to Q2 .

Q2 CHECK THE HEADLAMP SWITCH

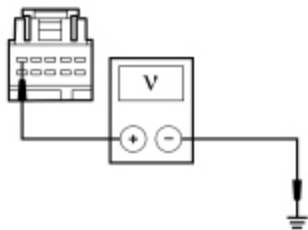
- Ignition OFF.
- Disconnect: Headlamp Switch [C205](#).
- Carry out the headlamp switch component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Does the headlamp switch pass the component test?

Yes	GO to Q3 .
No	INSTALL a new headlamp switch. REFER to Headlamp Switch in this section. TEST the system for normal operation.

Q3 CHECK FOR A VOLTAGE SIGNAL TO THE HEADLAMP SWITCH

- Ignition ON.
- Measure the voltage between the headlamp switch [C205](#) Pin 5, circuit CLF21 (GY/VT), harness side and ground.



• N0099713

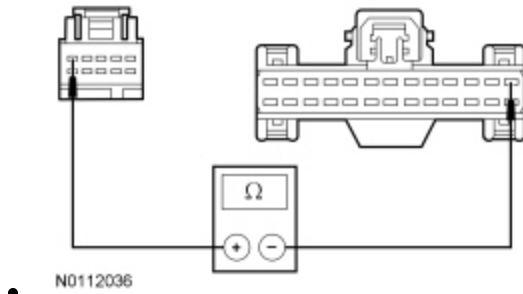
Is the voltage greater than 5 volts?

Yes	INSTALL a new headlamp switch. REFER to Headlamp Switch in this section. TEST the system for normal operation.
No	GO to Q4 .

Q4 CHECK THE FOG LAMP REQUEST INPUT CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: SCCM [C2414A](#) .

- Measure the resistance between the headlamp switch [C205](#) Pin 5, circuit CLF21 (GY/VT), harness side and the SCCM [C2414A](#) Pin 1, circuit CLF21 (GY/VT), harness side.



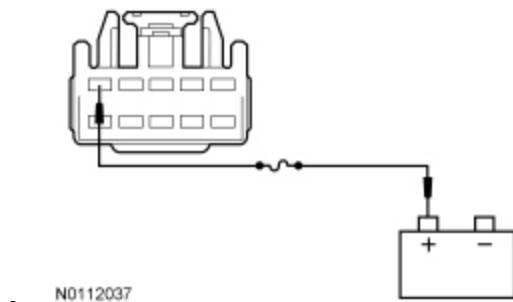
Is the resistance less than 5 ohms?

Yes	GO to Q6 .
No	REPAIR circuit CLF21 (GY/VT) for an open. TEST the system for normal operation.

Q5 BYPASS THE BCM

NOTE: If the fuse in the jumper wire fails, refer to the Wiring Diagrams manual to identify the possible causes of the circuit short.

- Ignition OFF.
- Disconnect: BCM [C2280E](#) .
- Connect a fused jumper wire between the BCM [C2280E](#) Pin 5, circuit CLF12 (BN/YE), harness side and battery positive.



Do the fog lamps illuminate?

Yes	REMOVE the jumper wire. GO to Q7 .
No	REMOVE the jumper wire. REPAIR circuit CLF12 (BN/YE) for an open. TEST the system for normal operation.

Q6 CHECK FOR CORRECT SCCM OPERATION

- Disconnect all the SCCM connectors.
- Check for:
 - corrosion
 - damaged pins

- pushed-out pins

- Connect all the **SCCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new SCCM . REFER to Section 211-05. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Q7 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test R: An Individual Fog Lamp Is Inoperative

Refer to Wiring Diagrams Cell [86](#) , Fog Lamps for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) monitors the fog lamp switch position by sending a voltage signal to the headlamp switch. When the fog lamp switch is engaged, the voltage signal is routed to ground indicating a request for the fog lamps.

The **SCCM** sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) , indicating the fog lamp switch position. When the **BCM** receives input from the **SCCM** indicating a request for the fog lamps, the **BCM** provides ground to the fog lamp relay (integrated into the **BCM**). When the fog lamp relay is energized, voltage is routed to the fog lamps. The fog lamps share a common ground circuit.

This pinpoint test is intended to diagnose the following:

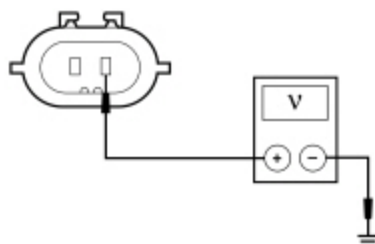
- Wiring, terminals or connectors

PINPOINT TEST R : AN INDIVIDUAL FOG LAMP IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

R1 CHECK FOR VOLTAGE TO THE FOG LAMP

- Ignition OFF.
- Disconnect: Inoperative Fog Lamp .
- Ignition ON.
- Place the headlamp switch in the PARKING LAMPS ON position and engage the fog lamp switch.
- Measure the voltage between the LH fog lamp [C152](#) Pin B, circuit CLF12 (BN/YE), harness side and ground; or between the RH fog lamp [C162](#) Pin B, circuit CLF12 (BN/YE), harness side and ground.



N0117368

Is the voltage greater than 10 volts?

Yes	REPAIR circuit GD120 (BK/GN) (LH fog lamp) or circuit GD123 (BK/GY) (RH fog lamp) for an open. TEST the system for normal operation.
No	REPAIR circuit CLF12 (BN/YE) for an open. TEST the system for normal operation.

Pinpoint Test S: The Fog Lamps Are On Continuously

Refer to Wiring Diagrams Cell [86](#) , Fog Lamps for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) monitors the fog lamp switch position by sending a voltage signal to the headlamp switch. When the fog lamp switch is engaged, the voltage signal is routed to ground indicating a request for the fog lamps.

The SCCM sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) , indicating the fog lamp switch position. When the BCM receives input from the SCCM indicating a request for the fog lamps, the BCM provides ground to the fog lamp relay (integrated into the BCM). When the fog lamp relay is energized, voltage is routed to the fog lamps.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Headlamp switch
- SCCM
- BCM

PINPOINT TEST S : THE FOG LAMPS ARE ON CONTINUOUSLY

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

S1 CHECK FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM fuse 47 (15A) .
- Ignition ON.
- Observe the fog lamps

Do the fog lamps continue to illuminate?

Yes	GO to S2 .
No	GO to S4 .

S2 CHECK THE BCM FOG LAMP INDICATOR OUTPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM [C2280A](#) .
- Ignition ON.
- Observe the fog lamps

Do the fog lamps continue to illuminate?

Yes	GO to S3 .
No	REPAIR circuit CLF12 (BN/YE) (between the BCM and the headlamp switch) for a short to voltage. TEST the system for normal operation.

S3 CHECK THE BCM FOG LAMP OUTPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM [C2280E](#) .
- Ignition ON.
- Observe the fog lamps

Do the fog lamps continue to illuminate?

Yes	REPAIR circuit CLF12 (BN/YE) (between the BCM and the fog lamps) for a short to voltage. TEST the system for normal operation.
No	GO to S7 .

S4 CHECK THE HEADLAMP SWITCH

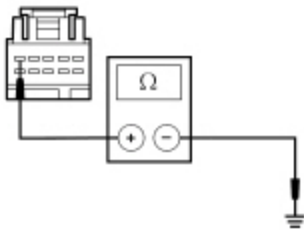
- Carry out the headlamp switch component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Does the headlamp switch pass the component test?

Yes	GO to S5 .
No	INSTALL a new headlamp switch. REFER to Headlamp Switch in this section. TEST the system for normal operation.

S5 CHECK THE FOG LAMP REQUEST INPUT CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: SCCM [C2414A](#) .
- Measure the resistance between the headlamp switch [C205](#) Pin 5, circuit CLF21 (GY/VT), harness side and ground.



• N0083977

Is the resistance greater than 10,000 ohms?

Yes	GO to S6 .
No	REPAIR circuit CLF21 (GY/VT) for a short to ground. TEST the system for normal operation.

S6 CHECK FOR CORRECT SCCM OPERATION

- Disconnect all the SCCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SCCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new SCCM . REFER to Section 211-05. TEST the system for normal operation.
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No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.
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S7 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test T: The Fog Lamps On Indicator Is Inoperative

Refer to Wiring Diagrams Cell [86](#) , Fog Lamps for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) , indicating the fog lamp switch position. When the **BCM** receives input from the **SCCM** indicating a request for the fog lamps, the **BCM** provides ground to the fog lamp relay (integrated into the **BCM**). When the fog lamp relay is energized, voltage is routed to the fog lamps and on a separate circuit, to the headlamp switch for the fog lamps on indicator.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Headlamp switch
- **BCM**

PINPOINT TEST T : THE FOG LAMPS ON INDICATOR IS INOPERATIVE

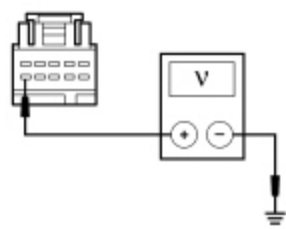
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

T1 CHECK THE HEADLAMP SWITCH

- Ignition OFF.
- Disconnect: Headlamp Switch [C205](#).
- Ignition ON.

- Enter the following diagnostic mode on the scan tool: **BCM DataLogger** .
- Select the **BCM fog lamps active command PID (FOG_LAMPS)**. Command the fog lamps on.
- **NOTE:** *Make sure the fog lamps are illuminated when taking a reading.*

Measure the voltage between the headlamp switch **C205** Pin 10, circuit CLF12 (BN/YE), harness side and ground.



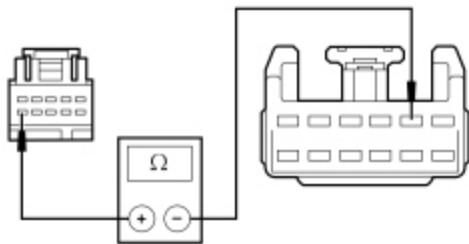
• N0083978

Is the voltage greater than 10 volts?

Yes	INSTALL a new headlamp switch. REFER to Headlamp Switch in this section. TEST the system for normal operation.
No	GO to T2 .

T2 CHECK THE BCM FOG LAMP INDICATOR OUTPUT CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: **BCM C2280A** .
- Measure the resistance between the headlamp switch **C205** Pin 10, circuit CLF12 (BN/YE), harness side and the **BCM C2280A** Pin 2, circuit CLF12 (BN/YE), harness side.



• N0112038

Is the resistance less than 5 ohms?

Yes	GO to T3 .
No	REPAIR circuit CLF12 (BN/YE) for an open. TEST the system for normal operation.

T3 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.

- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.